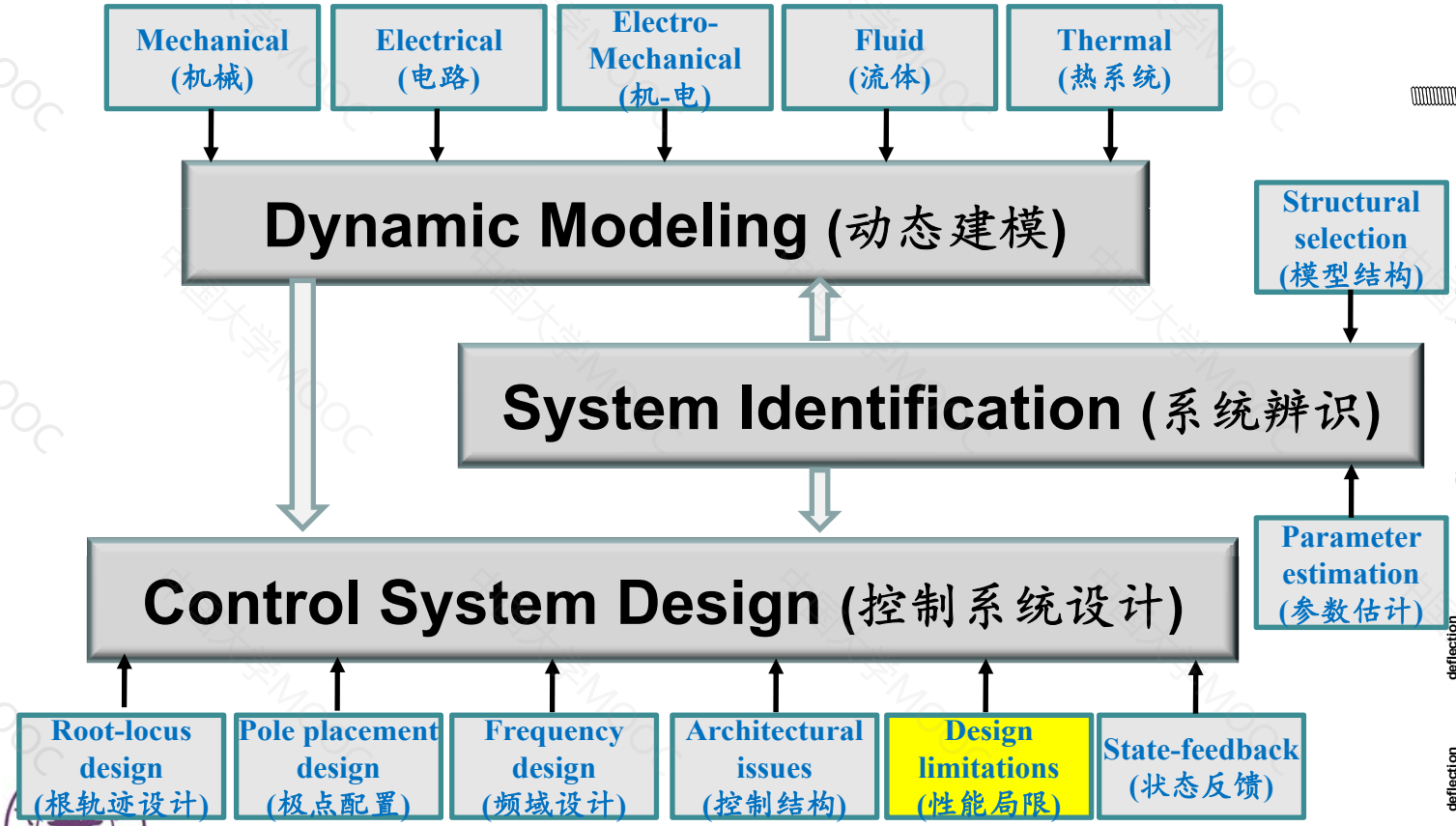
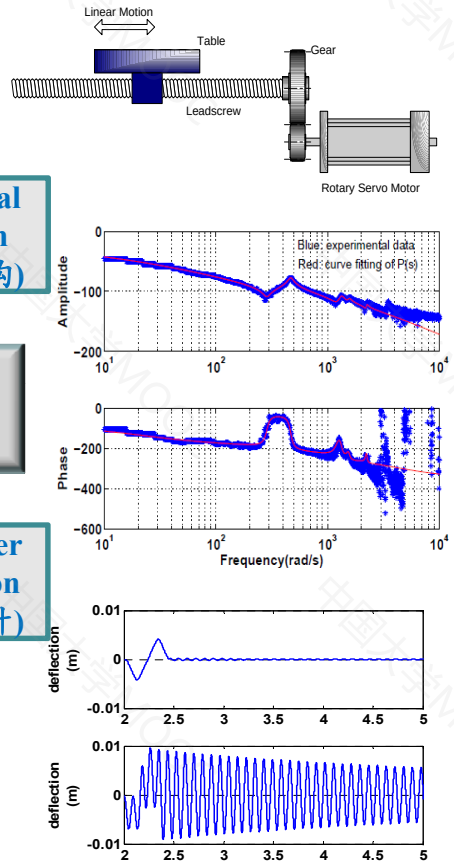
SISO Design Limitations
– Fundamental Design Trade-offs
(SISO设计局限性 – 基本的设计矛盾)

Main Contents of Course (课程主要内容)





Common Controller Design Issues

- **Closed-Loop Performance Specifications** (闭环性能指定)
  - **Command Following** (命令跟随)
    - Response speed, Steady-state error (响应速度、稳态误差) , …
  - **Disturbance Attenuation Properties** (干扰抑制特性)
    - Rejection of disturbance at steady-state, transient speed (稳态抗干扰能力、瞬时速度), …
  - **Practical Realizability** (实际可实现性)
    - Required control amplitude & control rate changes (所需控制幅度和控制量变化率), …
  - **Transient Response Quality** (瞬态响应质量)
    - Overshoot, if the system is open-loop unstable (超调，如果系统是开环不稳定的)
    - Undershoot, if the system is non-minimum phase (负调，如果系统是非最小相位的)
  - **Sensitivity of Design to Modeling Errors** (对建模误差的设计敏感性)

Fundamental SISO Design Trade-Offs

- **Disturbance Rejection Property** (抗干扰性)

$$Y_{dis}(s) = S(s)\left[d_o(s) + P(s)d_i(s)\right] \quad \& \quad S(s) = 1 - T(s)$$

For sinusoidal output disturbance (对于正弦输出干扰)

$$d_o(t) = A_d \sin(\omega t)$$

$$\Rightarrow \; y_{dis}(t) \rightarrow \; y_{ss}(t) = |S(j\omega)| \sin(\omega t + \angle S(j\omega))$$

Disturbances are rejected only at frequencies where (干扰仅在下述频段被抑制)

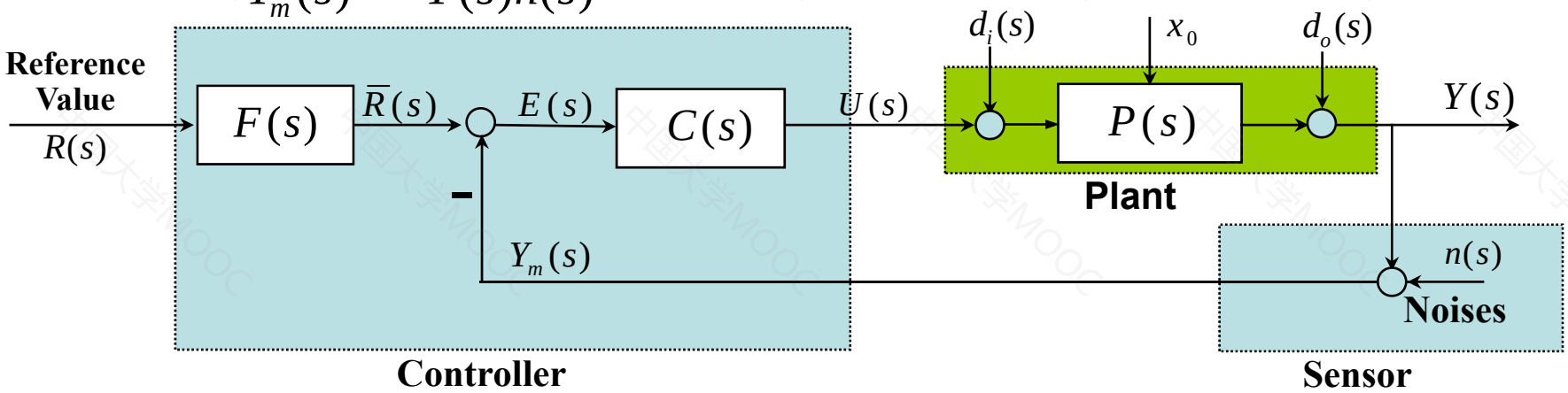
$$|S(j\omega)| \approx 0 \quad \text{or} \quad |T(j\omega)| \approx 1$$

To achieve acceptable performance in the presence of disturbances, it will generally be necessary to place a lower bound on the closed- loop bandwidth !
(为了在有干扰的情况下达到可接受的性能，通常需要设置闭环带宽的一个下界)

Fundamental SISO Design Trade-Offs

- **Measurement Noise Effect** (测量噪声的影响)

$$Y_m(s) = -T(s)n(s)$$



Measurement noise is rejected only at frequencies where (测量噪声被抑制的频段为)

$$|T(j\omega)| \approx 0 \qquad |S(j\omega)| \approx 1$$

Given the fact that noise is typically dominated by high frequencies, there will be an upper limit on the bandwidth of closed-loop set by measurement noise !
(考虑到噪声通常由高频主导的事实，测量噪声会需要设置闭环带宽的一个上界)

Fundamental SISO Design Trade-Offs

- **Control Effort** (控制量)

$$U(s) = S_u(s)\left[\bar{R}(s) - d_o(s) - n(s)\right] \quad \& \quad S_u(s) = T(s)/P(s)$$

$$S_u(s) = \frac{C(s)}{1+P(s)C(s)} = \frac{T(s)}{P(s)}, \quad \text{where} \quad T(s) = \frac{P(s)C(s)}{1+P(s)C(s)}$$

For frequency range that (在这些频率段) $|S_u(j\omega)| >> 1$,

Large control input may be needed at these frequencies
(在这些频率段需要较大的控制量)

Large control signals arise at frequencies where $|T(j\omega)| \approx 1$ but $|P(j\omega)| << 1$ which occurs when closed-loop is forced to be much more responsive than open-loop process (在 $|T(j\omega)| \approx 1$ 但 $|P(j\omega)| << 1$ 的频段需要较大的控制信号，这也是闭环控制而不是开环控制起主导的时候)

To avoid actuator saturation or slew rate problems, it will generally be necessary to place an upper limit on the closed- loop bandwidth!
(为避免执行器饱和或其变化率过大问题，通常有必要为闭环带宽设置一个上界)

Fundamental SISO Design Trade-Offs

- **Modeling Error Effect** (建模误差影响)

Robust Stability Theorem (鲁棒稳定性理论):

$$\left|T_n(j\omega)\right|\left|P_\Delta(j\omega)\right|<1, \quad \forall\omega$$

Design Constraint (设计约束):

$$S_n(j\omega)+T_n(j\omega)=1, \quad \forall\omega$$

$\Rightarrow$ For frequencies with significant modeling errors (在下列频段有较显著的建模误差):

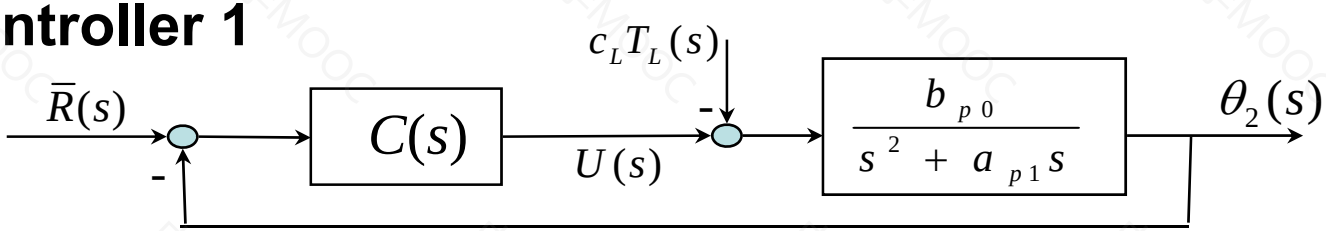
$$\left|P_\Delta(j\omega)\right|>>1 \Rightarrow \left|T_n(j\omega)\right|<<1, \left|S_n(j\omega)\right|\approx 1$$

Being responsive to reference changes and against disturbances at frequencies with significant modeling errors jeopardize stability; note also that MME $\left|P_\Delta(j\omega)\right|$ is normally significant at high frequencies (对参考量变化有响应和在有显著建模误差的频段上的抗干扰稳定性；注意MME通常在高频段比较显著)

To achieve robust stability and acceptable performance in the presence of modeling errors, it will generally be desirable to place an upper limit on closed-loop bandwidth !
(为获得鲁棒稳定性和有建模误差存在下可接受的性能，通常希望为闭环带宽设置一个上界)

PID Control of Rigid Belt ECP Emulator

- **PID Controller 1**



A proper PID controller has been previously synthesized to place CL poles at (之前已经设计一个正则PID控制器，配置闭环极点在） $p_{CL1,2d} = -20 \pm 20j$ & $p_{CL3,4d} = -40$

$$C(s) = K_P + \frac{K_I}{s} + \frac{K_D s}{\tau_D s + 1} = \frac{(K_D + K_P\tau_D)s^2 + (K_P + K_I\tau_D)s + K_I}{\tau_D s^2 + s}$$

$$\omega_n = 28\,\text{rad/s}$$

$$C(s) = \frac{b_{c2}s^2 + b_{c1}s + b_{c0}}{s^2 + a_{c1}s} = \frac{8.33s^2 + 205s + 2054}{s^2 + 117s}$$

$$S(s) = \frac{1}{1 + L(s)} = \frac{1}{1 + PC} = \frac{s^2(s + 117)(s + a_{c1})}{A_{CL}(s)}$$

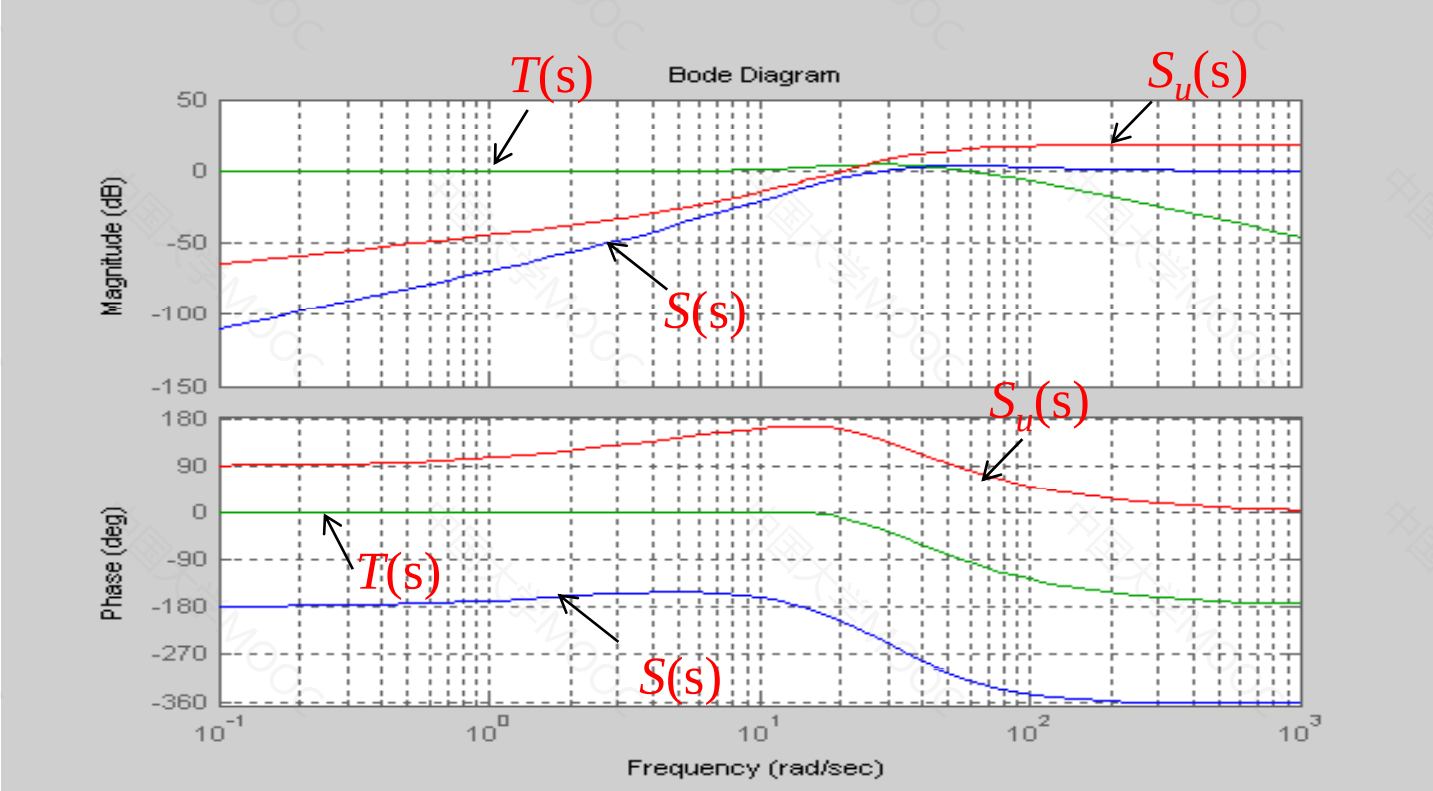
$$T(s) = \frac{PC}{1 + PC} = \frac{b_{p0}(8.33s^2 + 205s + 2054)}{A_{CL}(s)}$$

$$S_u(s) = \frac{C}{1 + PC} = \frac{s(s + a_{p1})(8.33s^2 + 205s + 2054)}{A_{CL}(s)}$$

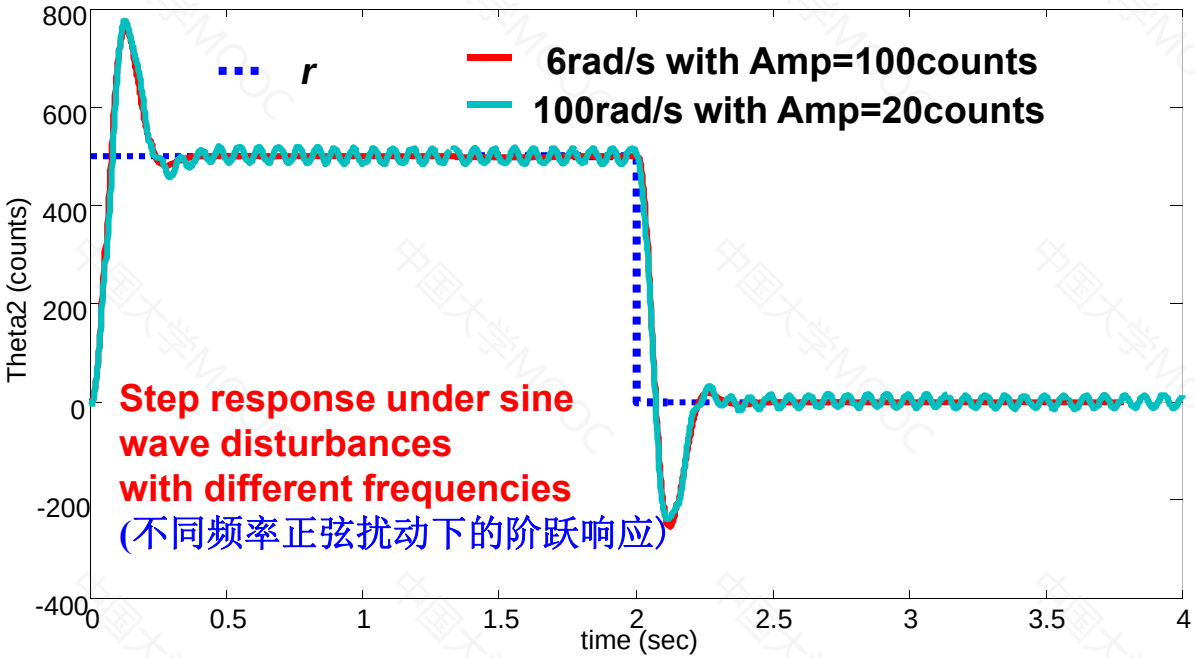
$$A_{CLd}(s) = [(s + 20)^2 + 20^2][s + 40]^2$$
$$= (s + 20 - 20j)(s + 20 + 20j)(s + 40)(s + 40)$$

PID Control of Rigid Belt ECP Emulator

- **Sensitivity Functions** (灵敏度函数)



Step Response under Sine Wave Disturbances

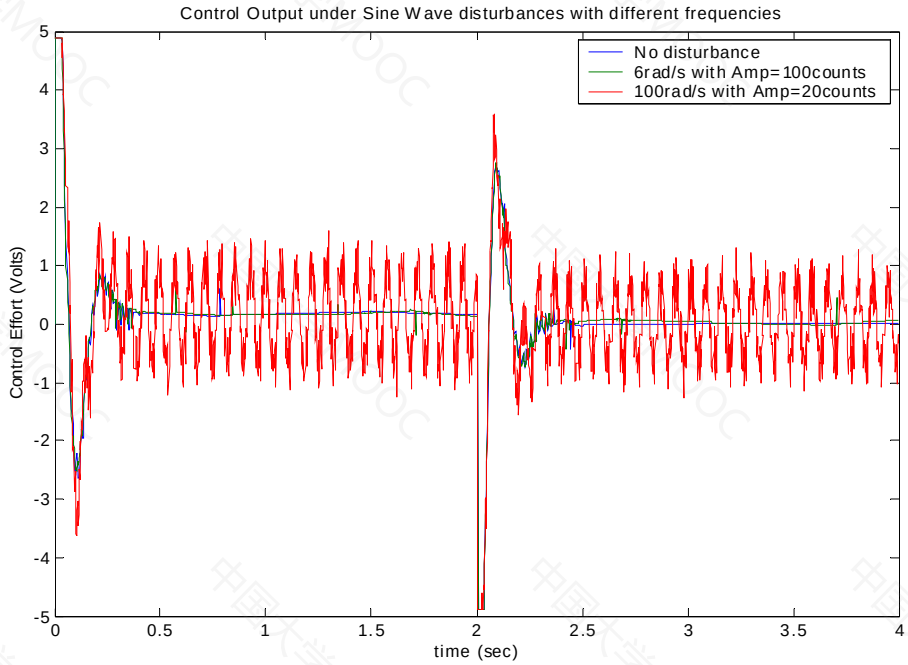


For 6 rad/s sine disturbance, the voltage applied to the disturbance motor has a magnitude of 0.065 V (equivalent to 100 counts at output side). 对于**(6rad/s的正弦干扰，作用在干扰电机上的电压是0.065V,等效于100个脉冲)**

For 100 rad/s sine disturbance, the voltage applied has a magnitude of 3.14 V (equivalent to 20 counts at output side). **(对于100rad/s的干扰，作用在干扰电机上的电压是3.14V，等效于20个脉冲)**

For disturbances within the CL bandwidth (e.g., 6 rad/sec), the system has very good disturbance rejection capability due to low sensitivity. For disturbances outside the CL bandwidth (e.g., 100 rad/sec), due to the low complementary sensitivity or sensitivity value close to 1, the effect of disturbance is not attenuated. **(对于闭环带宽内的干扰(如6 rad/sec)，由于灵敏度低，系统具有很好的抗干扰能力。对于闭环带宽以外的干扰(如100 rad/sec)，由于互补灵敏度较低或灵敏度值接近1，干扰影响没有减弱)**

PID Control Experimental Results



The large amplitude of sinusoidal control input at steady-state for sinusoidal output disturbance of 100 rad/sec is due to the large value of the control input sensitivity function at 100 rad/sec. **(由于控制输入灵敏度函数值在100 rad/sec时较大，所以100 rad/sec的正弦输出干扰下的稳态时控制输入的幅值较大)**

PID Control of Rigid Belt ECP Emulator

- **PID 2**

A proper PID controller that is synthesized to place CL poles at (一个正则PID控制器设计闭环极点在)

$$\Rightarrow \quad p_{CL1,2d} = -30 \pm 30j \quad \& \quad p_{CL3,4d} = -40$$

$$\omega_n = 30\sqrt{2} = 42\,\text{rad/s}, \zeta = 0.707$$

$$C(s) = \frac{12.39s^2 + 385s + 4620.6}{s(s+136.5)} \quad \text{or} \quad \begin{matrix} K_p = 2.57 & K_I = 33.85 \\ K_d = 0.072 & \tau_D = 0.0073 \end{matrix}$$

- **PID 3**

A proper PID controller that is synthesized to place CL poles at (一个正则PID控制器设计闭环极点在)

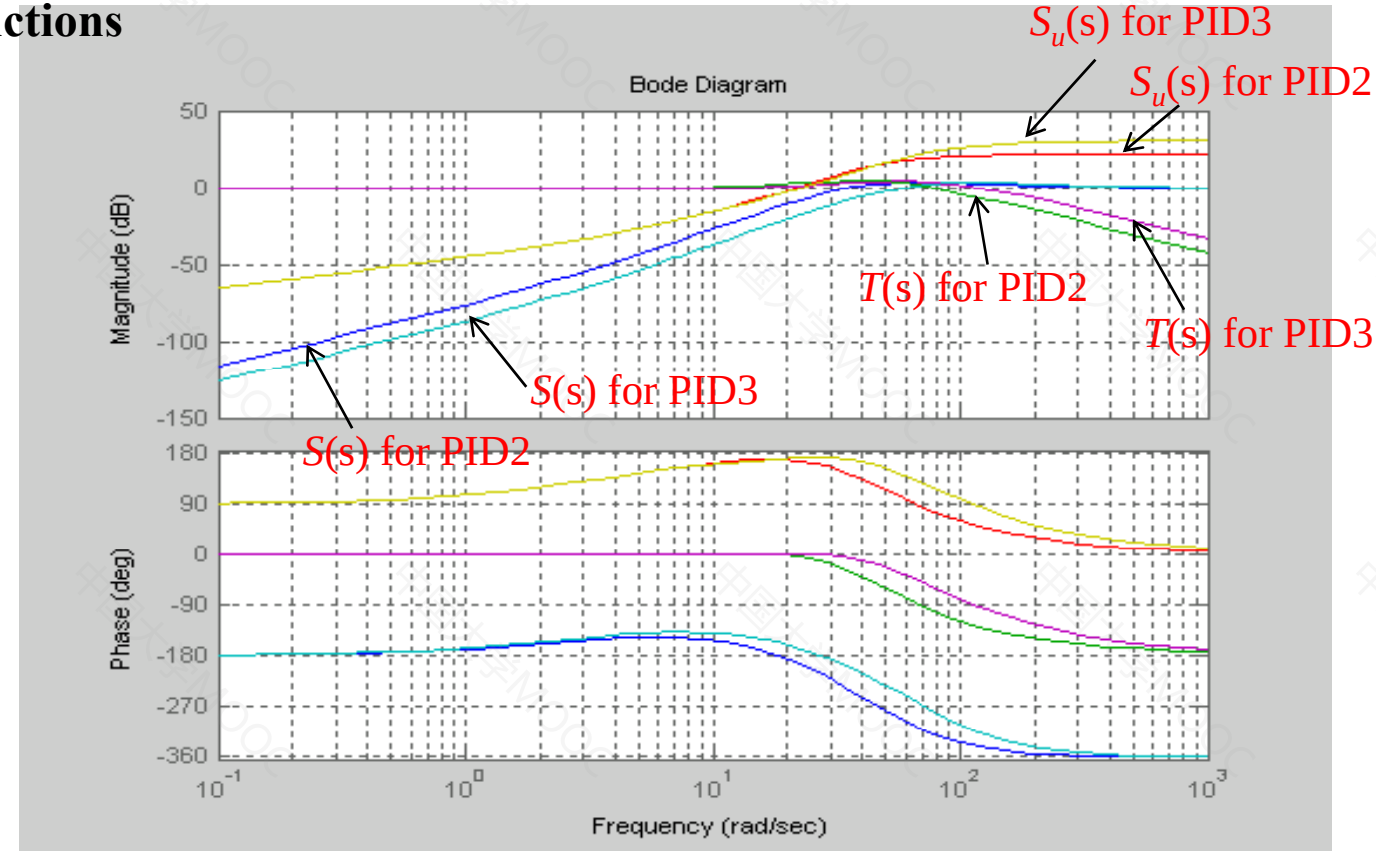
$$p_{CL1,2d} = -50 \pm 30j \quad \& \quad p_{CL3,4d} = -70$$

$$\Rightarrow \quad \omega_n = 58\,\text{rad/s}, \zeta = 0.857$$

$$C(s) = \frac{34.45s^2 + 1550s + 26729}{s(s+236.5)} \quad \text{or} \quad \begin{matrix} K_p = 6.075 & K_I = 113 \\ K_d = 0.12 & \tau_D = 0.004 \end{matrix}$$

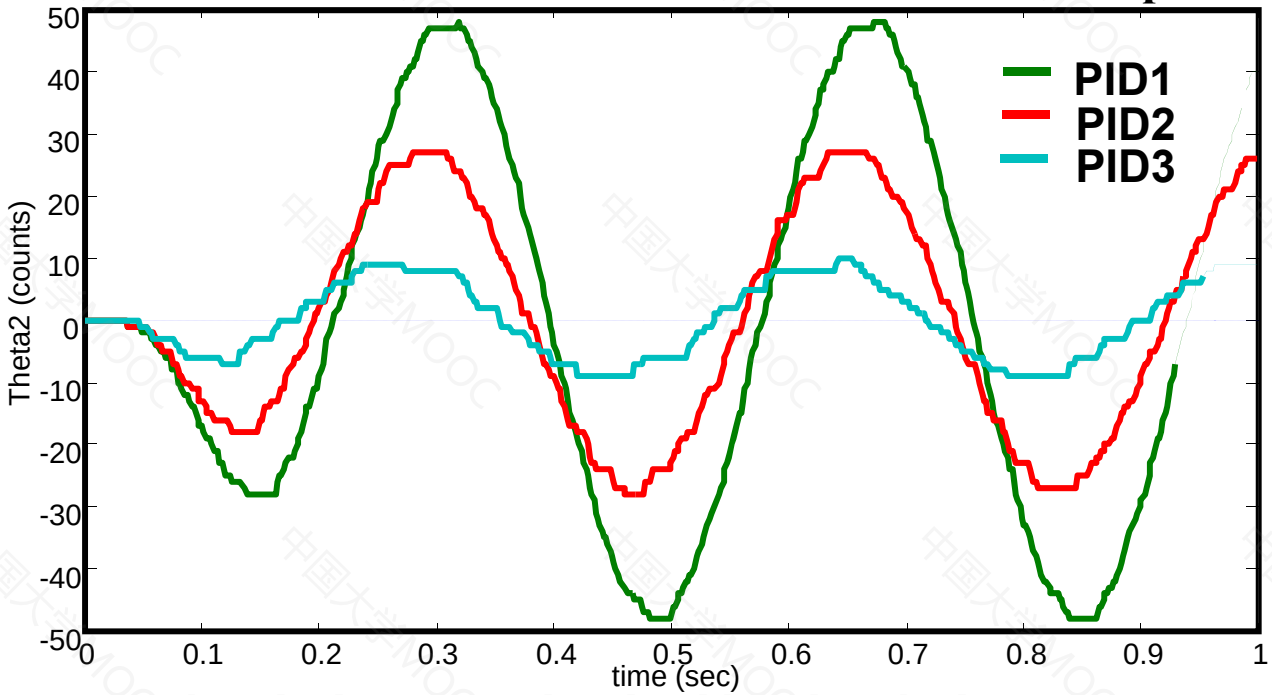
PID Control of Rigid Belt ECP Emulator

Sensitivity Functions



PID Control Experimental Results

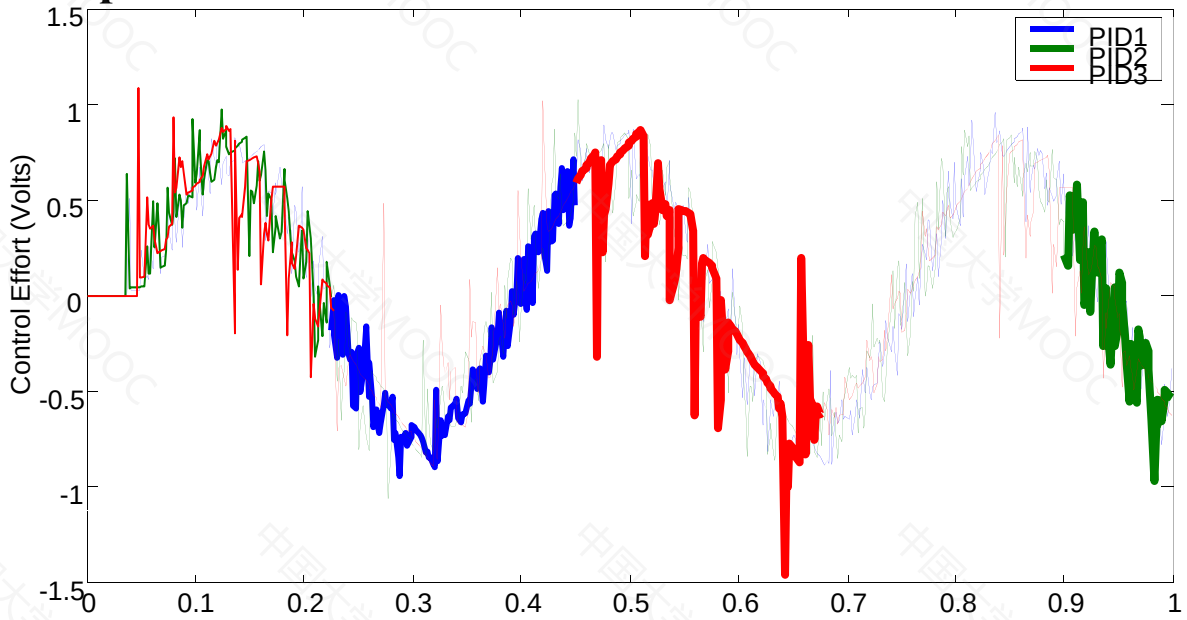
Response under Sine Wave disturbance of 20 rad/sec with Amp = 200 counts



As PID 3 has the largest CL bandwidth, it has the lowest sensitivity value at 20 rad/sec. As such, it has the best disturbance rejection capability. **(PID 3的闭环带宽最大，在ω= 20rad/sec的灵敏度值最低。因而具有最佳的抗干扰能力)**

PID Control Experimental Results

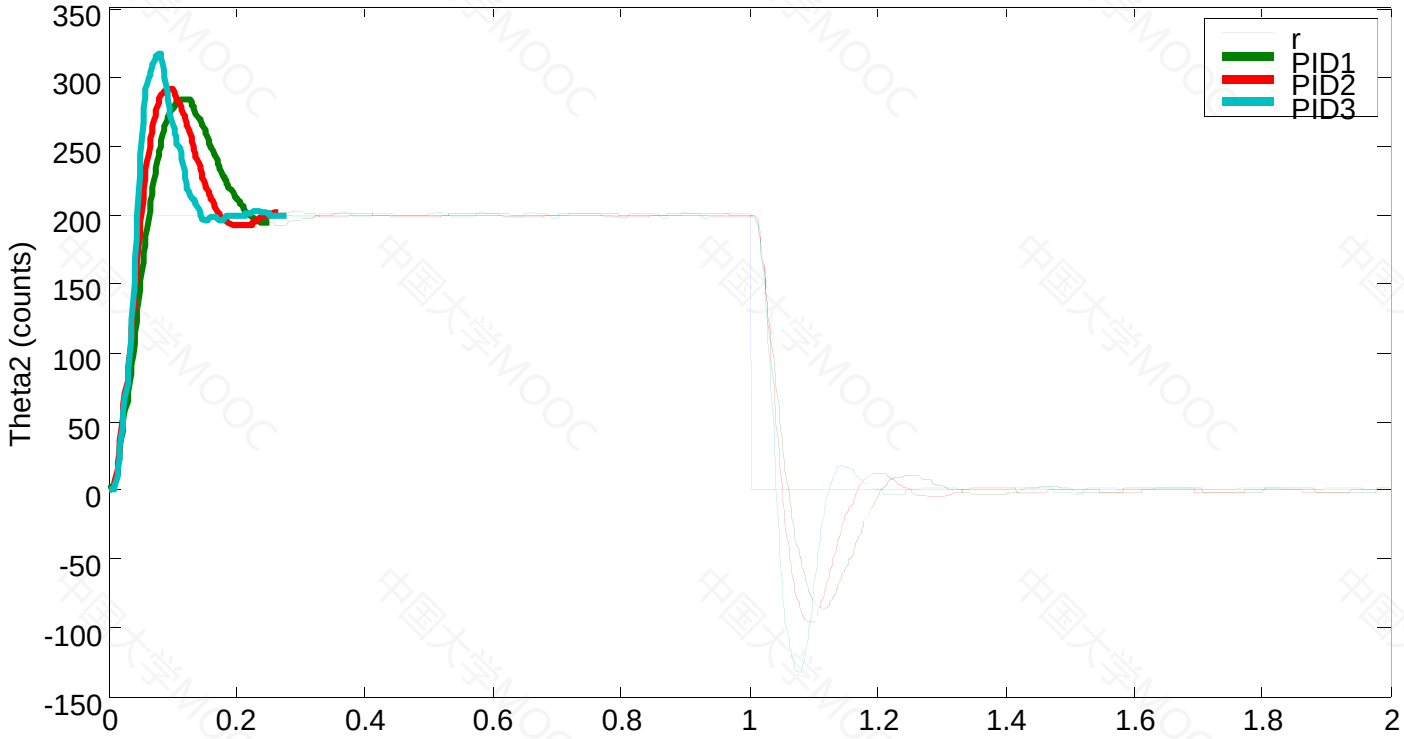
Control Output under Sine Wave disturbance of 20 rad/sec with Amp = 200 counts



As PID 3 has the largest CL bandwidth, it has the highest control input sensitivity value at all frequencies. As such, its control input tends to be choppier than PID1 and PID2 **(由于PID 3具有最大的闭环带宽,它在所有频段都有最高的控制输入灵敏度值。因此,它的控制量往往比PID1和PID2更波动)**

PID Control Experimental Results

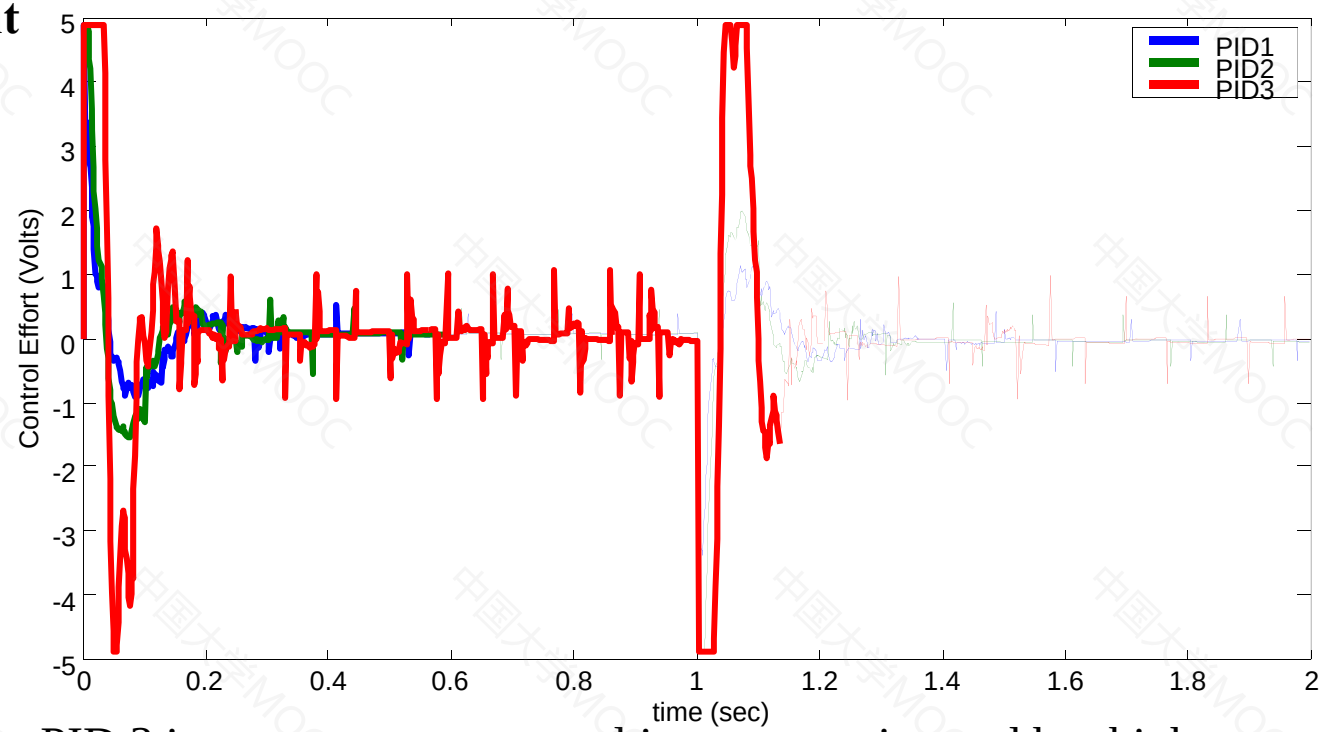
Step Response



As PID 3 has the largest CL bandwidth, it has the fastest response time **(由于PID 3的闭环带宽最大，所以它的响应时间最快).**

PID Control Experimental Results

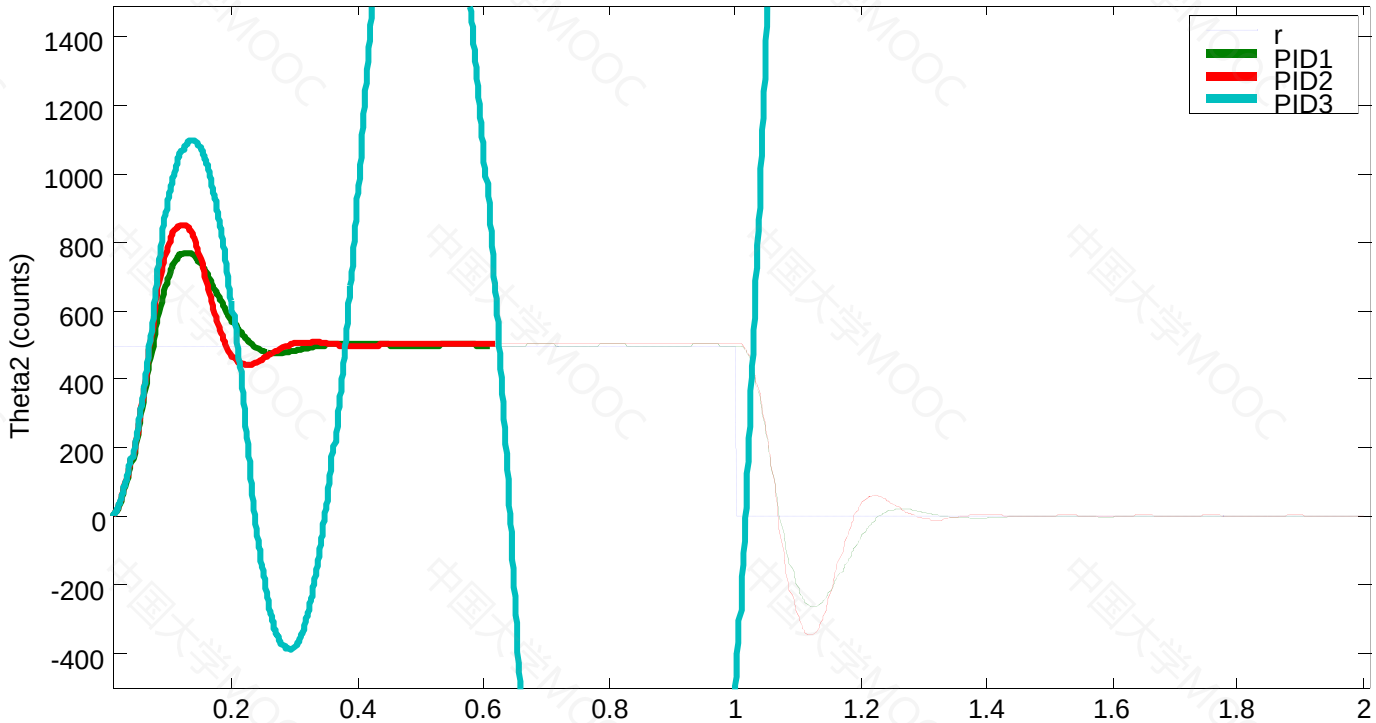
Control Input



However, PID 3 is more prone to control input saturation and has higher control input sensitivity (choppy control input) **(然而，PID 3更倾向于达到控制输入饱和且具有更高的控制输入灵敏度(波动的控制输入))**

PID Control Experimental Results

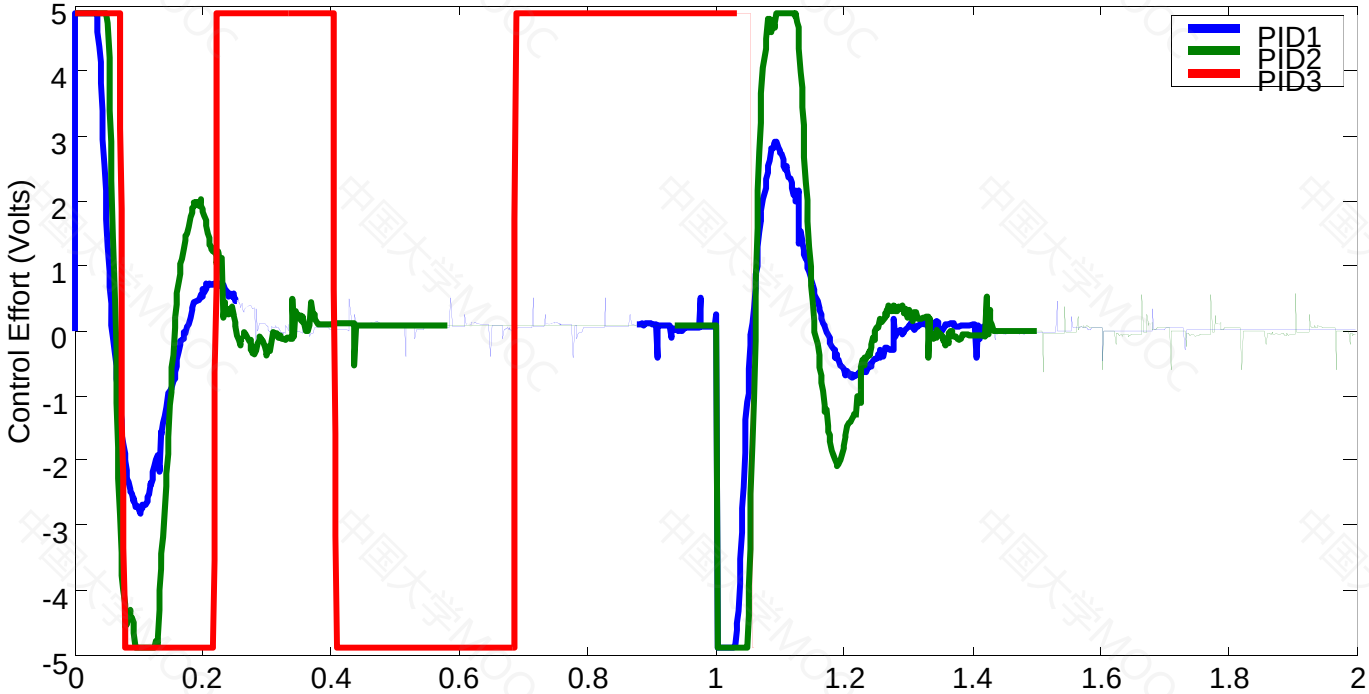
Step Response



In fact, for large reference input command, PID 3 has such a severe control input saturation problem that actually results in an unstable CL system (实际上，对于大的参考输入命令，**PID 3**存在严重的控制输入饱和问题，导致闭环系统不稳定)

PID Control Experimental Results

Control Input



The control input saturation problem of PID 3 and chopper control input of PID 2 when compared to PID 1 **(与PID 1对比，PID 3的控制输入饱和PID 2的控制振荡问题)**

Summary

- **Common Controller Design Issues** (常见控制器设计问题)
- **Fundamental Design Trade-offs** (基本设计平衡)
  - Disturbance Rejection Property (干扰抑制特性)
  - Measurement Noise Effect (测量噪声影响)
  - Control Effort (控制量)
  - Modeling Error Effect (建模误差影响)
- **Limitations due to Physical Components** (物理元件的限制)
  - Sensors (e.g., locations, accuracy, response time) (传感器(例如：位置、精度、响应时间)
  - Actuators (e.g., magnitude and rate saturation, resolution) (执行器(例如：幅度和饱和率、分辨率)